Introduction to Databases

Module Outcome:

- Say data vs database vs DBMS without mixing the three
- Pick relational vs document for a real app (CampusCart vs a chat log)
- Know why this course lives in MySQL tables, not a spreadsheet forever

Concepts in Focus

- Data
- Database
- Database Management System (DBMS)
- Advantages
- Types of Databases
- Relational Database
- Non-Relational Database

Data

Any sort of information that is stored is called **data**.

Examples:

1. Messages & multimedia on WhatsApp
2. Products and orders on Amazon
3. Contact details in telephone directory, etc.

In real applications, data includes fields like `firstname`, `lastname`, `date_of_birth`, `mobile_number`, and also content such as posts, images, likes, and shares.

Where Can We Store Data?

Before choosing a database, it helps to know the common storage options:

Option	Examples	Best for
Flat files	`.txt`, `.json`, `.yaml`, `.csv`, Excel	Small datasets, personal use, quick exports
Database	MySQL, PostgreSQL, MongoDB	Applications with many users, large data, and strict rules

Many beginners start with **Excel** because it is familiar. Excel works for small, personal datasets — but most software products move to a database as data and users grow.

Why Not Excel for Applications?

Limitation	What happens
Single flat file	All data sits in one sheet — hard to model relationships (e.g., customers and their orders)
Row limit	Excel supports roughly **1 million rows** per sheet — not enough for large apps
Concurrent users	Multiple users editing the same file at once causes conflicts and overwrites
Performance	Searching, filtering, and reporting slow down as data grows
Data integrity	Users can type wrong values (e.g., text in a salary column) — Excel does not enforce rules strictly

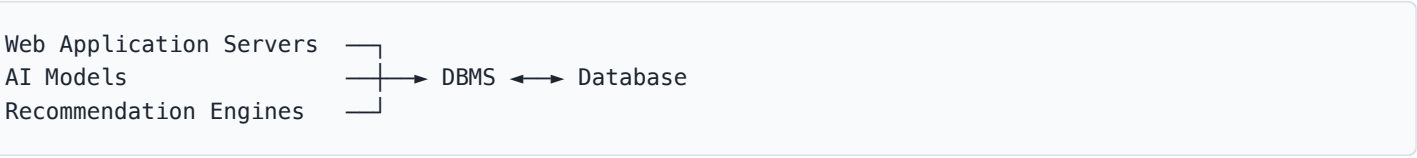
A **database** solves these problems: it stores related data in structured tables, handles many users at once, enforces data types and constraints, and stays fast at scale.

Database

An organized collection of data is called a **database**.

Database Management System (DBMS)

A software that is used to easily store and access data from the database in a secure way.



Advantages

- **Security:** Data is stored & maintained securely.
- **Ease of Use:** Provides simpler ways to create & update data at the rate it is generated and updated respectively.
- **Durability and Availability:** Durable and provides access to all the clients at any point in time.
- **Performance:** Quickly accessible to all the clients (applications and stakeholders).

Types of Databases

There are different types of databases based on how we organize the data:

Type	Description
Relational	Data organized in tables
Analytical	Central data with branching structures
Key-Value	Key mapped to value pairs

Type	Description
Column Family	Column-oriented storage
Graph	Nodes and edges
Document	Hierarchical document structure

Relational Database

In relational databases, the data is organized in the form of **tables**.

Tables, Rows, and Columns

A relational database stores data in **tables**. Understanding these three terms is essential before writing SQL.

Term	Description	Example (player table)
Table	A collection of related data organized in rows and columns	`player`
Column	A vertical field that stores one type of data (attribute)	`name`, `age`, `score`
Row	A horizontal record representing one entry (tuple)	`('Virat', 32, 50)`

```
player table
┌──────────┬─────┬───────┐
│ name     │ age │ score │  ← columns (attributes)
├──────────┼─────┼───────┤
│ Virat    │ 32  │ 50    │  ← row 1
│ Rakesh   │ 39  │ 35    │  ← row 2
│ Sai      │ 47  │ 38    │  ← row 3
└──────────┴─────┴───────┘
```

DBMS vs RDBMS

	DBMS	RDBMS
Full form	Database Management System	Relational Database Management System
Data format	Files, objects, documents, graphs, etc.	Tables with rows and columns
Relationships	Limited or no formal relationships	Enforces relationships via keys
SQL support	May or may not support SQL	Uses SQL as the standard language
Examples	MongoDB, Redis, Cassandra	MySQL, PostgreSQL, Oracle, SQL Server

> **Note:** All RDBMS are DBMS, but not all DBMS are RDBMS. This course focuses on **RDBMS using MySQL**.

Non-Relational Database

Graph, Key-Value, Column Family, and Document — these four types are commonly referred to as **non-relational databases**.

> **Note:**
> - Choice of database depends on our requirements.
> - Relational database is the most commonly used database.

Relational DBMS

A Relational DBMS is a DBMS designed specifically for relational databases. Relational databases organize the data in the form of tables.

Examples: MySQL, Oracle, PostgreSQL, SQLite, SQL Server, IBM DB2, etc.

> **Note:** This course uses **MySQL** as the primary RDBMS. See MySQL Setup and Connection for installation.

Non-Relational DBMS

A Non-Relational DBMS is a DBMS designed specifically for non-relational databases. Non-relational databases store the data in a "non-tabular form".

Examples: Elasticsearch, CouchDB, DynamoDB, MongoDB, Cassandra, Redis, etc.